

Total No. of Questions : 6]

SEAT No. :

P 5518

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BE/Insem./Oct.-94

B.E. (IT) (Semester - I)

Soft Computing (Elective - I)

(2012 Pattern)

Time : 1 Hour]

[Max. Marks :30

Instructions to the candidates:

- 1) *Figures to the right indicate full marks.*
- 2) *Assume suitable data, if necessary.*

- Q1)** a) Compare Hard Computing Vs Soft Computing. [4]
b) What is expert system? What are performance goals of next generation expert systems? [6]

OR

- Q2)** a) How semantic networks are used for knowledge representation and processing? [4]
b) List and comment on Architectures of knowledge based systems. [6]

- Q3)** a) Implement OR function using McCulloch-Pitts Perceptron model. (Consider Binary data) [4]
b) List and characterize various activation functions in neural networks. [6]

OR

- Q4)** a) Explain with example non-linearly separable pattern classification. [6]
b) Obtain output of Neuron Y for inputs $[x_1, x_2, x_3] = [0.8, 0.6, 0.4]$ and the weights are $[w_1, w_2, w_3] = [0.1, 0.3, -0.2]$ with bias $b = 0.35$ and its input as 1. Use binary sigmoidal activation function. [4]

- Q5)** a) Draw and explain architecture of Kohonen's Self Organization Feature Map (KSOFM). [6]
b) Give limitations and applications of Hopfield network. [4]

OR

- Q6)** a) State and explain Hebb's learning Rule for competitive learning. [4]
b) Describe need and properties of ART network. [6]

